

Name: _____

Homework 3 | Math 341 | Cruz Godar

Due Wednesday of Week 4 at the start of class

Complete the following problems and submit them as a pdf to Canvas. 8 points are awarded for thoroughly attempting every problem, and I'll select three problems to grade on correctness for 4 points each. Enough work should be shown that there is no question about the mathematical process used to obtain your answers.

Section 3

1. Let $\vec{v} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$ and $\vec{w} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$.

- a) Draw $\vec{v}$ and $\vec{w}$ in the plane.
- b) Draw $2\vec{v} + \vec{w}$ geometrically and verify that it matches the algebraic definition (i.e. adding the entries of $\vec{v}$ and $\vec{w}$).
- c) Draw a vector linearly dependent with $\vec{v}$, but linearly independent with $\vec{w}$.
- d) Draw a vector linearly dependent with both $\vec{v}$ and $\vec{w}$, but not with either of them alone.

In problems 2–5, determine if the vectors are linearly dependent or independent. If they are dependent, find a linear combination equal to $\vec{0}$.

2. $\vec{v}_1 = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$, $\vec{v}_2 = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$.

3. $\vec{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, $\vec{v}_2 = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$, $\vec{v}_3 = \begin{bmatrix} 1 \\ 4 \\ 4 \end{bmatrix}$.

4. $\vec{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$, $\vec{v}_2 = \begin{bmatrix} 2 \\ 0 \\ 3 \end{bmatrix}$, $\vec{v}_3 = \begin{bmatrix} 4 \\ -1 \\ 2 \end{bmatrix}$

$$5. \vec{v}_1 = \begin{bmatrix} 1 \\ 5 \\ 6 \\ -10 \\ \frac{17}{2} \end{bmatrix}, \vec{v}_2 = \begin{bmatrix} 3 \\ 1 \\ 7 \\ -100 \\ 0 \end{bmatrix}, \vec{v}_3 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \vec{v}_4 = \begin{bmatrix} 4 \\ 1 \\ 2 \\ 0 \\ 89 \end{bmatrix}.$$

In problems 6–7, express $\vec{v}$ as a linear combination of the $\vec{u}_i$ or show it's impossible.

$$6. \vec{v} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \vec{u}_1 = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}, \vec{u}_2 = \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix}, \vec{u}_3 = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}.$$

$$7. \vec{v} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \vec{u}_1 = \begin{bmatrix} -1 \\ 3 \\ 0 \end{bmatrix}, \vec{u}_2 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \vec{u}_3 = \begin{bmatrix} 2 \\ 6 \\ 3 \end{bmatrix}.$$

$$8. \text{ Let } \vec{w}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \vec{w}_2 = \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix}, \vec{w}_3 = \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix}.$$

- Show that $\vec{w}_1$, $\vec{w}_2$, and $\vec{w}_3$ are linearly dependent.
- Show that just $\vec{w}_1$ and $\vec{w}_2$ on their own are linearly independent. (Hint: You should be able to modify the last step in the previous part to get this result without starting over).
- Using the previous two parts, write a sentence explaining why

$$\text{span}\{\vec{w}_1, \vec{w}_2\} = \text{span}\{\vec{w}_1, \vec{w}_2, \vec{w}_3\}.$$

- The span of $\vec{w}_1$ and $\vec{w}_2$ is the set consisting of all $\vec{w} = c_1\vec{w}_1 + c_2\vec{w}_2$. Renaming $c_1 = u$ and $c_2 = v$, write the generic vector $\vec{w}$ in the span. Your answer should depend on u and v .
- [Math3D](#) is a capable 3D grapher that can handle parametric surfaces. Open the linked example and replace the span expression with the one you found in the previous part — if all went well, you should

see the three vectors lying *in* that plane. This plane is the span, and the fact that the three vectors are contained in a two-dimensional surface is their linear dependence.

Section 4

9. Linear transformations are related to typical linear functions like $y = mx + b$, but they're not quite the same. For example, the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = 2x + 3$ is a linear function, but not a linear transformation. Pick two inputs a and b and show that $f(a + b) \neq f(a) + f(b)$.

In problems 10–12, find the matrix for the linear transformation T and use it to evaluate $T(\vec{v})$.

$$10. T \left(\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \right) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, T \left(\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right) = \begin{bmatrix} 3 \\ 4 \end{bmatrix}, T \left(\begin{bmatrix} 0 \\ 0 \\ 3 \end{bmatrix} \right) = \begin{bmatrix} 3 \\ 1 \end{bmatrix}, \text{ and } \vec{v} = \begin{bmatrix} 2 \\ 3 \\ 2 \end{bmatrix}.$$

$$11. T \left(\begin{bmatrix} 1 \\ 1 \end{bmatrix} \right) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, T \left(\begin{bmatrix} 1 \\ -1 \end{bmatrix} \right) = \begin{bmatrix} -1 \\ 1 \end{bmatrix}, \text{ and } \vec{v} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}.$$

$$12. T \left(\begin{bmatrix} 0 \\ 1 \end{bmatrix} \right) = 2, T \left(\begin{bmatrix} 1 \\ 3 \end{bmatrix} \right) = 7, \text{ and } \vec{v} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

13. Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be the linear transformation given by

$$T \left(\begin{bmatrix} x \\ y \\ z \end{bmatrix} \right) = \begin{bmatrix} 2x + 3y \\ x + 2y \\ 3x + z \end{bmatrix}.$$

a) Express T as a 3×3 matrix A .

b) Find A^{-1} .

c) Write a linear transformation S whose matrix is A^{-1} .

- d) Since $A^{-1}A = I$ and matrix multiplication is equivalent to function composition, we should expect $S \circ T = id$, the identity function. Evaluate $S \circ T$ as functions by using the output of T as the input to S , and show this is in fact the case.